Victoria Chen
Rashi Jeeda
GitHub: github.com/rjeeda/20440_TARA

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Uncovering Drivers of Ocean Microbiome Diversity

Introduction

Earth's oceans are a deep reservoir of microbial diversity. Oceanic microbes play crucial roles in geochemical and biological processes, including carbon and nutrient cycling[1]. These processes have significant implications for global climate cycles and ecological dynamics. However, the environmental factors driving changes in oceanic microbial communities have not been well-studied. Ocean microbiomes are influenced by many conditions, including temperature, nutrient content, and light exposure. Quantifying the impact of these environmental factors on microbial dynamics in the ocean is important, as it can help guide understanding of the implications of human activity on biodiversity. For example, human activity is responsible for recent increases in global temperatures, as well as ocean acidification and deoxygenation[2]. Additionally, this type of analysis is important for better understanding how to maintain synthetic ocean environments, such as aquariums[3]. Understanding how marine microbiomes are related to different environmental factors can help conservationists and animal caretakers ensure their habitats represent natural environments as closely as possible.

The *Tara* Oceans expedition currently stands as the most comprehensive global survey of ocean microbiome composition to date[4]. By collecting, analyzing, and sequencing ocean water samples from various depths across the globe, this study offers resources for a holistic understanding of the ocean's microbial ecosystem. Though other ocean genomics projects have been launched, *Tara* has gone a step further by integrating genetic and functional diversity with environmental metadata at a global scale and at multiple depths. This enables systems-level analysis of ocean microbial community dynamics[5]. In total, the *Tara* expedition collected 139 prokaryote-enriched samples across the globe (Figure S1A). For each sample, the following metadata was recorded: biome, environmental feature, region, province, salinity, latitude, longitude, depth, oxygen, temperature, nitrogen dioxide, phosphate, and nitrate concentrations.

Previous analysis of the 16S rRNA sequencing-based taxonomic abundance data in Sunagawa et al.'s original *Tara* Oceans study found that temperature strongly influences microbiome composition in the sunlit epipelagic ocean later[1]. Depth was also identified as a significant factor in stratifying oceanic microbial communities based on taxonomic abundances[1]. However, less work has been done to explore which other environmental factors correlate strongly with overall oceanic diversity, rather than with surface-level dynamics. Even a recent 2022 paper describing the *Tara* Oceans work acknowledged that our understanding of marine ecosystems in the surface (top 200 meters) of the ocean far exceeds that of deeper ecosystems[4].

Moreover, though the *Tara* Oceans data includes both functional and taxonomic composition, Sunagawa et al.'s analysis predominantly focused on taxonomic composition. A deeper dive into functional abundances and how they compare to taxonomic abundances would be helpful for gaining new insights into ocean ecology. Previous studies on the human gut microbiome have noted that function appears to be more highly conserved across samples than across taxa, which could suggest that function is more robust across communities, since individual taxa come and go[6,7]. Here, robustness is defined as resistance to changes in community composition across environmental conditions. The species composition of microbial communities is reported to experience frequent fluctuations. This may naturally lead to variations in functional profile as taxa with genes encoding important functions move in and out of a community. Understanding the magnitude of such shifts can help inform ecological insights, such as whether all taxa tend to share important functions. Currently, it is reported that a systematic study of the line between community composition and robustness is generally lacking in microbiome studies[8].

Overall, we are interested to know (1) if the same environmental factors are correlated to variation in both taxa and function throughout the ocean, (2) how the relationship between taxa and functional composition changes across the range of these most important environmental

factors, and (3) what the most important taxa or functional genes are that drive these associations with the environment.

Answering (1) will give us insight into the overall ecology of the ocean and inform what environmental factors necessitate community adaptation. We approach this by conducting a PCoA and investigating which environmental metadata, if any, separate the samples and whether these are different for taxa and function. We also create a Random Forest classifier to predict these environmental variables from taxa and function composition, allowing us to determine which compositional data is more informative. Investigating (2) will provide insight into how functional genes are distributed in ocean ecosystems. We will compare richness values for taxa and function in the same environments. If we see that functional richness does not change much even when taxonomic richness does, this would imply that many taxa can carry out these functions. Finally, answering (3) will inform whether specific genes and taxa may have unique adaptations to certain conditions in the ocean. We will be able to infer the genes and taxa most relevant to different environmental conditions by looking into the features of top importance in our Random Forest models.

We hypothesize that taxonomical and functional composition in the ocean microbiome may shift differently in response to the same environmental factors, as seen in the human gut microbiome[8]. Specifically, based on previous literature, we hypothesize that functional abundances will be more stable than taxonomical abundances across environmental changes, since this is true in the gut microbiome[7]. Many of the variables we are investigating are related to nutrient concentrations, and it is known that deeper layers of the ocean have less organic matter than the surface[9]. Thus, we also hypothesize that the taxa and functions most associated with different environmental conditions will be those related to metabolism.

Methods

Data collection. All data used in this analysis were downloaded from the *Tara* Oceans companion website (ocean-microbiome.embl.de/companion.html). The files used were:

- *Companion Website Tables W1-W8*, a spreadsheet file containing multiple types of metadata
- *annotated 16S OTU count table*, containing OTU counts summarized at different taxonomic levels for prokaryote-enriched samples (N=139)
- *eggNOG OG profile (prokaryotic only)*, a table of eggNOG gene family abundances for prokaryote-enriched samples (N=139)

To map orthologous groups to their functional categories, the following files were downloaded from the eggNOG version 3 database (eggnog.embl.de/version_3.0/downloads.html):

- *COG.funccat.txt.gz*, clusters of orthologous groups and their functional categories
- *NOG.funccat.txt.gz*, non-supervised orthologous groups and their functional categories

Data processing. Clusters of orthologous groups (COGs), collections of proteins inferred to be orthologs, belonging to the ocean core were filtered out according to Companion Table W7. We focused on non-core functions, since these exhibited a far larger variance in relative abundance (mean $\sigma^2 = 1.87 \times 10^{-7}$ for non-core orthologous groups (OGs) vs 5.21×10^{-9} for core OGs) across ocean samples, as stated by the TARA Oceans analysis paper. Core functions were too numerous to efficiently run our model, so non-core functions were chosen since they were tractable and informative based on this variance. Operational taxonomic unit (OTU) and OG abundance counts were normalized in each table such that the sum of abundances for any given sample was 1. Relative abundance counts were also calculated at the phylum and gene functional category levels. These relative abundance counts were used for all further analysis.

Principal coordinate analysis Principal coordinate analysis (PCoA) was performed using the 'pcoa' method of scikit-bio. Two PCoA ordinations were made for samples, one based on taxa abundances and one on functional abundances. PCoA was chosen over principal component

analysis (PCA) since PCA uses Euclidean distance, which is not relevant to ecological data. In the ocean microbiome, the presence/absence of certain features could have non-linear effects on community composition. To account for this, PCoA was performed on Bray-Curtis distances between samples. Bray-Curtis is commonly used in microbiome analyses and in ecological community contexts[10]. Additionally, PCoA is known to perform better than PCA on sparse data, such as microbiome data with many low-abundance microbes or those which are missing from many samples[11]. To evaluate which environmental factors best separated points in the PCoA ordination, Pearson correlations were calculated between each environmental variable and the top most informative PCs for each PCoA. A p-value was also calculated for each correlation.

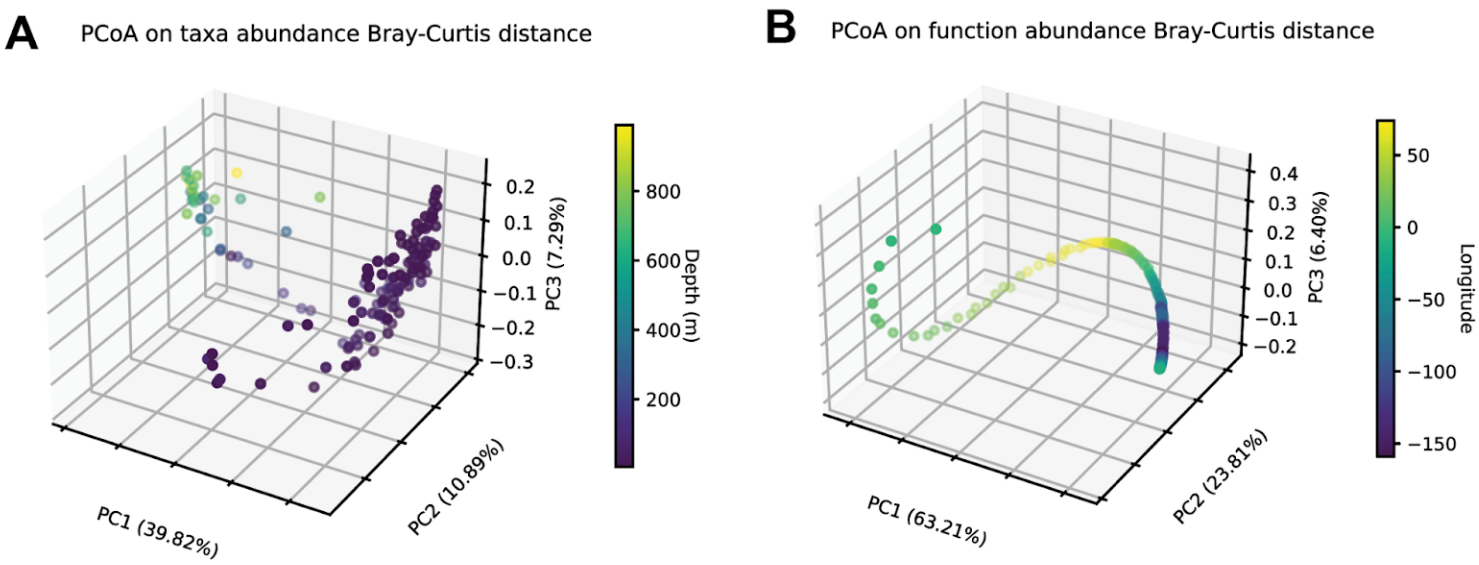
Random forest modeling. scikit-learn's 'RandomForestRegressor' was used to fit continuous variables, and 'RandomForestClassifier' was used to fit categorical variables. Each Random Forest was set to include 500 trees, and was fit using 85% of the input dataset for training. The remaining 15% of the dataset was used for validation of model performance. Each Random Forest model was re-fit and tested 20 times with a different train-test split for cross-validation, and the average accuracy scores of the 20 runs was recorded. R^2 values were used to score the regressors, and scikit-learn's built in accuracy metric was used for the classifiers.

Richness analysis. Richness analysis was performed on both taxa and function across the most important features determined by PCoA and our Random Forest models. Taxonomic richness is defined as the number of unique taxa in a sample and feature richness as the number of unique non-core COGs in a sample. Values were normalized separately for taxa and function by dividing all values by the highest seen for any sample. Normalization was done to make taxa and function richness values comparable to each other, as they are known to be on scales that differ by an order of magnitude. The analysis consisted of a comparison of taxa richness and functional richness over all depth layers and temperature levels in the oceans. The variances of taxonomic and functional richness in each environmental condition were compared using an F test.

Results

PCoA to test if different environmental factors correlate with taxonomy and function

We performed a principal coordinate analysis (PCoA) to identify the most important metadata features driving community differences based on taxa and function. The results of our PCoA analysis on taxonomic versus functional abundances show that the two do seem to be driven by different environmental factors.



C

Data	PC	Env. Variable	Corr. Coeff.	p-value
taxa	**1**	**Depth (m)**	**0.8643697**	**P<0.001**
taxa	1	Nitrates (pmol/L)	0.8241144	P<0.001
taxa	1	Temperature (°C)	0.773033	P<0.001
taxa	**2**	**Temperature (°C)**	**0.4514803**	**P<0.001**
taxa	2	Oxygen (pmol/kg)	0.3986219	P<0.001
taxa	2	Nitrates (pmol/L)	0.2086313	P=0.027

D

Data	PC	Env. Variable	Corr. Coeff.	p-value
func	**1**	**Longitude**	**0.7577026**	**P<0.001**
func	1	Salinity (PSU)	0.664327	P<0.001
func	1	Latitude	0.3088036	P<0.001
func	**2**	**Latitude**	**0.5339193**	**P<0.001**
func	2	Longitude	0.408724	P<0.001
func	2	Salinity (PSU)	0.1546703	P=0.072

Figure 1. PCoA on taxa and functional abundances and corresponding correlations to environmental metadata. **(A)** PCoA results on taxa abundance, colored by depth. **(B)** PCoA results on functional abundance, colored by longitude. **(C)** Table detailing the Pearson correlation coefficient of the top three correlated environmental variables with each of PC1 and PC2 for taxa, along with p-value of Pearson test. **(D)** Similar to C, but for function.

In the PCoA performed on taxa abundance, PC1 explained 40% of the variance, while PC2 explained 11% of the variance (Fig. 1A). PC1 was most highly correlated with sample depth values, while PC2 was most highly correlated with sample temperature values. In the

PCoA performed on functional abundance, PC1 explained 63% of the variance and PC2 explained 24% of the variance (Fig. 1B). PC1 was most highly correlated with sample longitude values (Fig. 1C), while PC2 was most highly correlated with sample latitude values (Fig. 1D). The PCoA plots are colored by the variable that most highly correlated with PC1 in each case, as this was the PC that explained the most variance in each. Visually, we see that the data are indeed separated by these variables.

Random forest modeling to test if different environmental factors correlate with taxonomy and function and which is more informative

In order to see which environmental variables could be well-predicted by supervised machine learning, we created several Random Forest models. We fit continuous variables (latitude, longitude, depth, temperature, salinity, oxygen concentration, nitrate concentration, nitrogen dioxide concentration, and phosphate concentration) using Random Forest regressors, and fit categorical variables (environmental feature, biome, and region) using Random Forest classifiers. To determine what type of input data yielded the best results, separate sets of models were fit using OTU (taxonomic) and OG (functional) abundances. We also fit models to broader definitions of these categories - phylogenetic and gene functional category abundances.

In comparing the mean accuracy scores for each model, we saw that depth and temperature (Figure 2A) were the best performing models across all types of input data. Salinity and NO_2 concentration, which did not show much variation across oceanic zones (Supplemental Figure 1C), had the worst model performance. Overall, models trained on taxonomic inputs (OTU and phylogenetic abundances) performed better than functional inputs (OG and functional category abundances) (Figure 2C). Additionally, grouping the data into higher levels (phylogenetic and functional category abundances) lowered model performance as well (Figure 2C). This contradicted previous work which found that lower-level taxonomies, in particular

phylum level abundances, produced more accurate Random Forest models[12]. This suggests that the most informative taxonomy level varies with environment and application.

The best Random Forest classifier models performed worse than the best regression models, likely due to the sparse amount of data for certain categories such as non-surface samples and non-trades samples (Figure 2B, 2C).

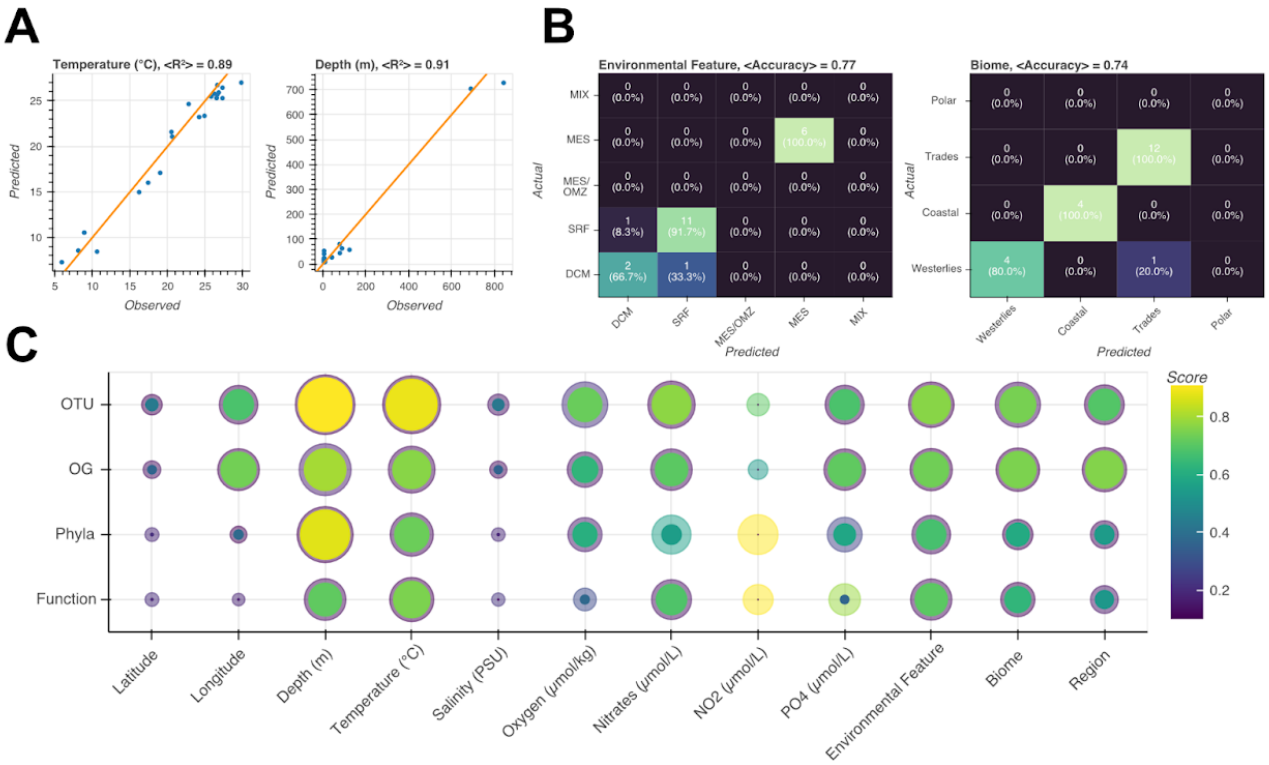


Figure 2. (A) Representative scatter plots of best performing cross-validation run for Random Forest models predicting depth and temperature trained on OTU relative abundances. **(B)** Representative confusion matrices of best performing cross-validation run for Random Forest models predicting environmental feature and biome trained on OTU relative abundances. **(C)** Bubble heatmap plot of average accuracy score (R^2 for regression models, first 9 columns and scikit-learn built-in Random Forest accuracy score for classifier models, last 3 columns) for each model based on 20 cross-validation runs. Larger, brighter circles correspond to more accurate models. Each circle is surrounded by a ring indicating standard deviation, with larger, brighter rings corresponding to higher standard deviations.

Richness analysis to investigate how taxa and function vary across conditions

The results of our richness analysis found differences in how richness changes throughout environmental conditions for taxa and function. First, we saw that both taxonomic

and functional richness increased with depth. We saw the highest taxonomic and functional richness values for the MES environmental feature (Figure 3A), which constitutes the deepest sampled layer of the ocean.

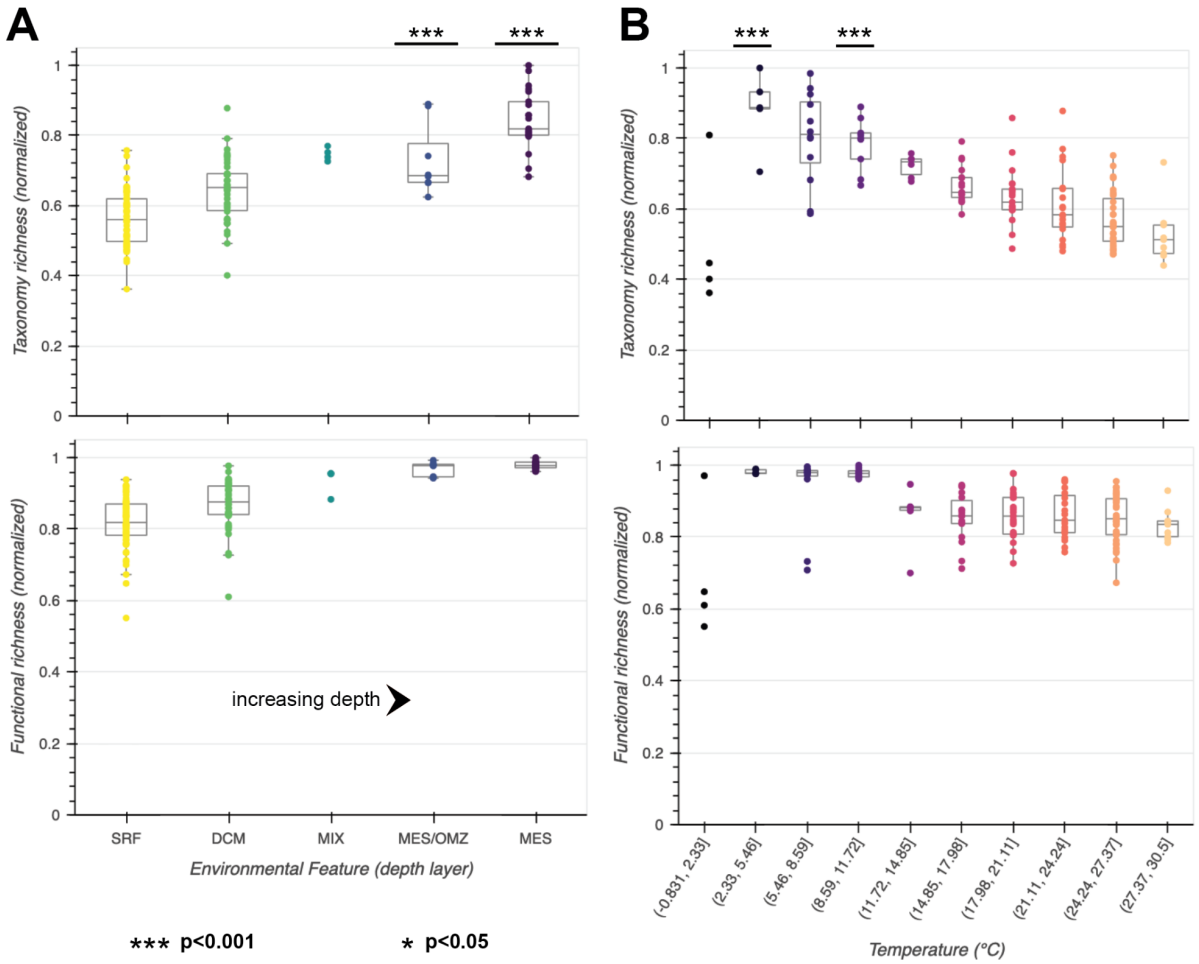


Figure 3. Richness analysis on taxa and function composition throughout changing environments. Statistical significance is calculated for difference in variance between taxa and functional richnesses for each category (F-test). **(A)** Taxonomy richness of all samples, grouped by environmental feature (depth layer in the ocean that the sample came from). **(B)** Functional richness of all samples, grouped by temperature range that samples came from.

For both taxonomic and functional richness, there was a steady increase in median richness from surface level to deeper ocean. When comparing the variance in taxonomic and functional richness for each condition, we saw that the difference in variance was significant to $p < 0.001$ (using an F test) for the MES and MES/OMZ environmental features, with functional richness having a much smaller variance than taxonomic richness.

Temperature was another important factor implicated in both our PCoA analysis and Random Forest models. Thus, we also investigated the change in richness across the temperature gradient of the ocean. We saw a steady decrease in taxonomic richness from colder parts of the ocean to warmer ones (Figure 3B, top plot). However, it seems that functional richness is stable, remaining high from -0.831 degrees to 11.72 degrees, followed by a decrease and another region of stability from 14.85 degrees and higher (Figure 3B, lower plot). When comparing the variance in taxonomic and functional richness for each condition, we saw that the difference in variance was significant to $p < 0.001$ for temperature ranges (2.33, 5.46] and (8.59, 11.72].

<u>Random Forest feature importances to investigate particular markers for specific conditions</u>

To look for particular biological markers for specific environmental conditions, such as certain depth zones, temperature ranges, or biomes, we looked at the top feature importances from each Random Forest model.

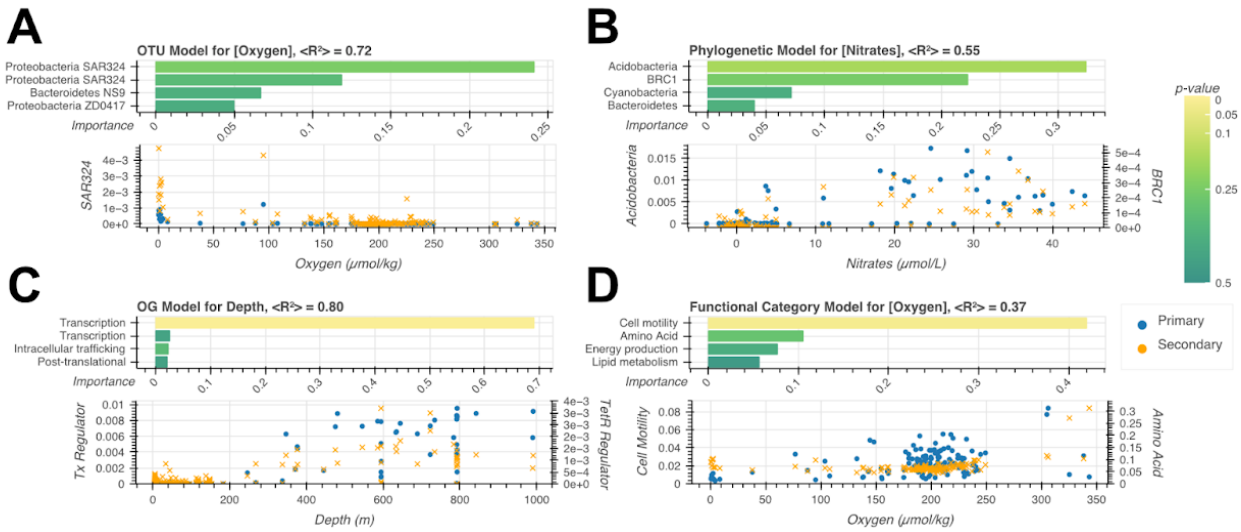


Figure 4. Top feature importances of interest from the best performing cross-validation run from Random Forest regression models for **(A)** oxygen concentration, trained on relative OTU abundances; **(B)** nitrate concentration, trained on relative phylogenetic abundances; **(C)** depth, trained on relative OG abundances; **(D)** oxygen concentration, trained on relative functional category abundances. The relative abundance versus the modeled variable is plotted for the top two (primary and secondary) features. Feature importance scores were calculated as the mean of the Gini impurity decrease for each tree. p-values were calculated from the Z-score for each feature using the cumulative distribution function (CDF) of the standard normal distribution.

Although most of the top features did not have significant p-values, investigating them still revealed several interesting relationships. Many of these were “all-or-nothing” patterns. For instance, the SAR324 clade ($p = 0.24$, $p = 0.33$) was enriched only in anaerobic conditions, with relative abundance close to 0% at non-zero oxygen concentrations (Figure 4A). Similarly, Acidobacteria ($p = 0.17$) were depleted at nitrate concentrations below 10 μmol/L (Figure 4B). COG4742[13] ($p = 0.009$), a predicted transcriptional regulator, and NOG40373 ($p = 0.43$), a TetR transcriptional regulator[14], were only found at low depths below 200 meters (Figure 4C). Cell motility genes ($p = 0.02$) were depleted in anaerobic conditions, and enriched at higher oxygen concentrations (Figure 4D).

Discussion

<u>(1) Different environmental factors may correlate with taxonomy and function</u>

The results of our PCoA ordination showed that PC1 of taxonomy composition was most highly correlated with depth, while PC1 of functional composition was most highly correlated with longitude. PC2 in each case was also correlated with different environmental factors. For taxonomy, the most important variables were generally depth, nitrates, and temperature. For function, the most important features were latitude, longitude, and salinity. This could indicate that differences in taxonomic composition are more related to changes in nutrient availability (which are associated with depth and nitrates), while changes in function are more related to geographic specialization (associated with latitude and longitude) and robustness to salinity.

However, the results of our Random Forest model showed that even on the basis of functional abundance, depth and temperature were the most reliably predicted variables. The discrepancy between this result and the PCoA results may be due to there being more variation for depth and temperature in our data set than for other variables. Our top-performing models being depth and temperature suggest that the variations in taxonomic and functional compositions across these environmental conditions were informative enough to accurately predict these conditions. This is consistent with the findings from Sunagawa et al.’s original *Tara*

Oceans study, which identified temperature as the main environmental driver for community composition. Additionally, Sunagawa et al. discuss the vertical stratification of microbe and virus compositions due to the high variance in parameters such as temperature, oxygen, and nutrient concentration with changing depths. Salinity and NO_2 are not thought to vary much across the ocean and are thus unlikely to drive microbial community concentration; this is likely why model performance for these variables was low, as there were no informative taxonomic or functional units to lower the Gini impurity.

Additionally, we aimed to use the Random Forest approach to investigate whether functional abundances were more correlated to environmental variables than taxa abundances. However, our results suggest that as-is, contrary to our expectations, functional abundances are a less reliable predictor overall than the conventionally-used taxonomic abundances. Further work should be done to investigate potential avenues for improving the predictive power of functional abundances, since this type of modeling would provide valuable insights into the metabolic capabilities and ecological roles of microbial communities.

<u>(2) Richness analysis reveals that function has less variance than taxa in some environments</u>

We observed that both taxonomic and functional richness increased with depth. This supports our hypothesis that deeper layers of the ocean may require more specialization in response to the higher pressures and sparser nutrient concentrations. We also saw that the variance in functional richness was smaller than that for taxonomic richness at the highest depths, which implies that while many taxa are present at these depths, there is a narrower range of non-core functions. One explanation for this could be that a multitude of functions are necessary for survival in the deep ocean, and that even in different geographic parts of the deep ocean, these functional needs do not vary much.

Generally, our results show that functional composition is robust to taxonomic diversity in deep ocean layers as well as at low temperatures. Even though taxonomic richness varies

greatly in these environmental conditions, functional richness maintains little variance in the same conditions. One explanation for this could be that in these environments, key highly abundant taxa perform a majority of functions. Another explanation could be that many taxa tend to share the same non-core functions. This warrants further exploration of specific gene content in the genomes of different taxa using metagenomic sequences. Going deeper into this question could provide insight as to how genes are moving between taxa, and whether ecological rates of gene transfer are different in low depth and temperature environments as opposed to shallow or warmer ones.

(3) Random forest feature importances reveal particular markers for specific conditions.

Our taxonomic model for temperature attained slightly higher cross-validation scores (best $R^2 = 0.96$, mean $R^2 = 0.89$, Figure 2A) than Sunagawa et al.'s Elastic Net model ($R^2 = 0.86$). This suggests that Random Forest modeling was able to uncover additional relationships (such as complex non-linear relationships and feature interactions) that were not captured using Elastic Net. Additionally, Random Forest provided an intuitive way to explore important features via averaging of the decrease in Gini impurity. Investigating these top feature importances in our models revealed some trends of interest. One of these was the SAR324 clade, which was important in estimating oxygen concentration. This clade was previously reported to possess machinery for both aerobic and anaerobic respiration[15,16]. However, our findings are indicative of a preference towards an anaerobic[17] lifestyle. This prompts further investigation into the factors that might make this advantageous, and why a nearly complete respiratory chain is maintained in the genome despite their abundance in oxygen minimum zones.

Additionally, our analysis identified two transcriptional regulators, COG4742 and NOG40373, that were exclusively present at lower depths. As expected, these regulators are associated with species inhabiting deep ocean environments (especially oxygen-depleted zones), such as Nitrosopumilus[18] and Methanobacteria[19]. This suggests that these genes could be optimized for oxygen minimum zones. However, for many of these species, no clear

connection with depth has been established in the literature. Future work could focus on determining whether the evolution of these genes is driven mainly by adaptation to low depth or low oxygen environments, or if these factors are functionally interconnected.

Our analysis also revealed a depletion of cell motility genes in oxygen-deprived conditions. This could be because anaerobic respiration is less efficient than aerobic respiration, driving microbes to conserve energy by reducing active motility in search of scarce nutrients. Instead, these microorganisms might employ alternative strategies, such as improving stationary nutrient uptake or forming biofilms on symbiotes.

One noteworthy observation was the increased abundance of Acidobacteria in environments with higher nitrate concentrations. This is particularly interesting given that, in soil ecosystems, Acidobacteria are known to possess nitrogen-processing genes but not the complete pathway for full denitrification to dinitrogen gas[20,21]. However, we hypothesize that in the ocean, Acidobacteria may indeed possess this full denitrification pathway, as denitrification plays an important role in marine ecology[22]. This hypothesis is supported by the fact that anaerobic conditions are required for full denitrification[23]; although Acidobacteria are predominantly aerobic in soil environments[21], preliminary evidence points to an anaerobic lifestyle in ocean environments[24].

Conclusion and Future Directions

We investigated the correlation of several environmental factors with oceanic diversity, and compared the predictive power of taxonomic and functional abundances. We observed an increase in richness with depth, suggesting that a higher level of specialization is required for the competitive environments found in deeper ocean layers. Functional composition was found to be less varied than taxonomic composition, indicating that either key species perform a majority of functions, or that many species possess similar non-core functions.

We successfully used Random Forest modeling to predict depth and temperature, which are known drivers of microbial community composition. However, contrary to our initial

hypothesis, functional abundances proved to be less reliable than taxonomic abundances in predicting environmental variables. In the future, several avenues for improving the predictive power of functional abundances could be explored. Network analysis could be used to pinpoint specific associations, and different types of omics data could be incorporated to gain a more comprehensive view of community function dynamics. Based on these results, new features, such as ratios of specific OGs, could be developed which may help model performance.

Investigation of top feature importances revealed interesting trends, such as the clear presence or absence of certain members in specific conditions. This analysis proved to be a good hypothesis generation technique, as our findings here prompt further exploration into the contributors behind these observations. For example, analyzing the involvement of understudied ocean Acidobacteria in key nitrogen cycle processes such as nitrogen fixation and denitrification by BLASTing their genomes for nitrite reductase (*nir)* genes could help give valuable insight into the role of microbes in marine ecology.

Better understanding the relationship between taxonomic and functional composition in the ocean and how they respond to different environmental factors can help further predictive environmental monitoring. This can also help guide intervention strategies for ocean conservation. Additionally, identifying microbes and functions associated with specific environmental niches can lead to the discovery of novel bioactive compounds. For example, extremophiles may produce enzymes that are suited for use in industrial processes which occur under high temperature or pressure. Overall, studying marine microbial communities, their functions, and ecological roles is important for global climate cycles and marine ecological dynamics. Our findings here have confirmed previous findings and generated new hypotheses about potential important members and functions in the ocean microbiome.

Supplement

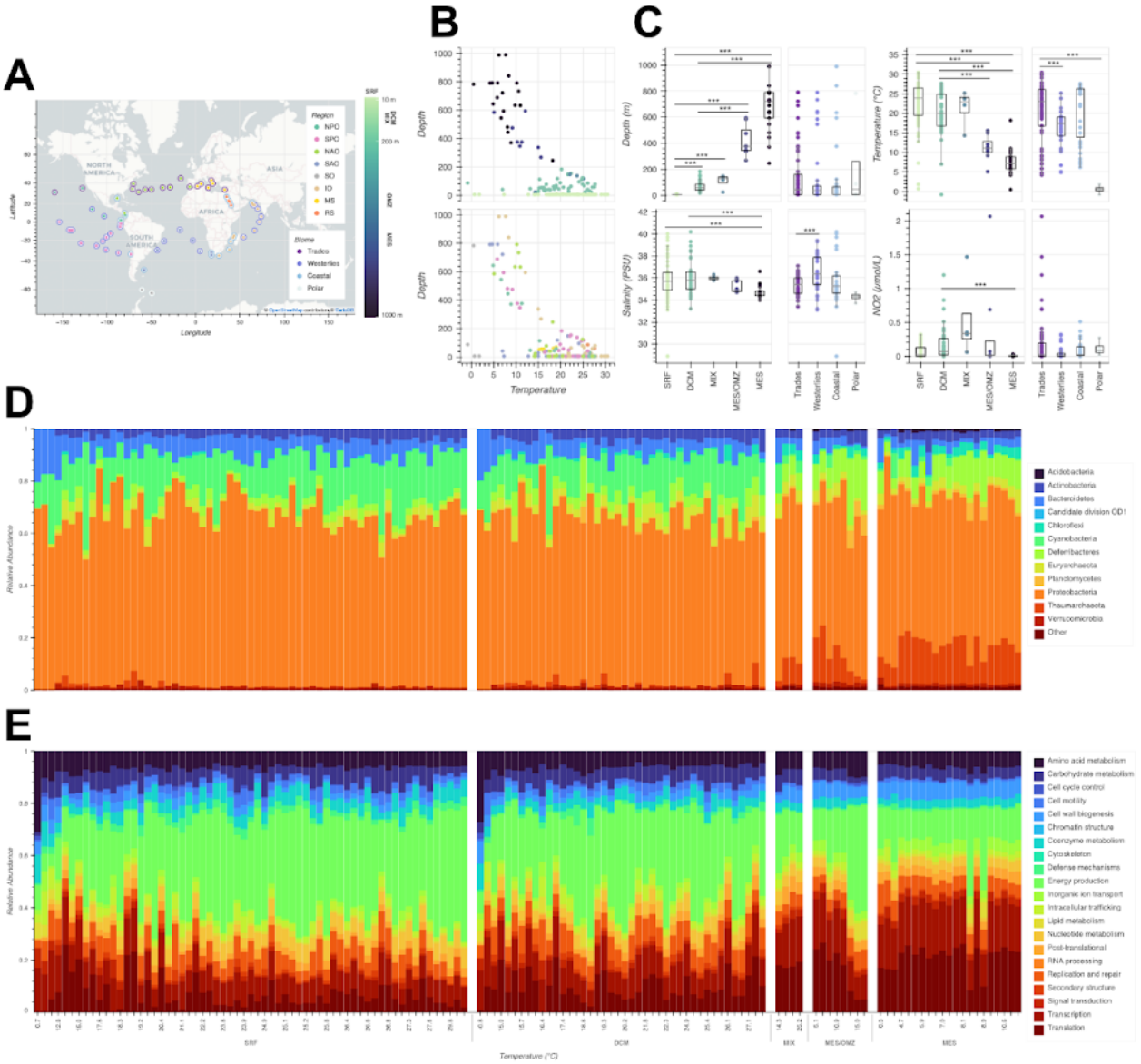


Figure S1. Overview of *Tara* Oceans dataset. **(A)** Map of sampling locations for the 139 prokaryote-enriched samples analyzed. Regions included the North Pacific Ocean (NPO), South Pacific Ocean (SPO), North Atlantic Ocean (NAO), South Atlantic Ocean (SAO), Southern Ocean (SO), Indian Ocean (IO), Mediterranean Sea (MS), and Red Sea (RS). Colorbar shows the depth intervals of each environmental feature, including the surface (SRF), deep chlorophyll maximum (DCM), mixed layer (MIX), oxygen minimal zone of the mesopelagic zone (OMZ/MES), and the mesopelagic zone (MES). **(B)** Scatterplot of depth versus temperature showing separation by environmental feature, but not by region. **(C)** Barplots of depth, temperature, and salinity and NO2 concentration. Significant ($p < 0.001$) differences in distributions between environmental feature and biome, as calculated by a two-sided Mann-Whitney U test, are represented by triple-starred (***) bars. **(D)** Relative abundances of phyla found in the *Tara* Oceans dataset. The x-axis is sorted by environmental feature and temperature. **(E)** Relative abundances of gene functional categories found in the *Tara* Oceans dataset. The x-axis is sorted by environmental feature and temperature.

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